

UNIT 4 · WEEKS 5–6

Processes and memory

📅 Weeks 5–6 · dates to be confirmed <> Lab TD3 (solved + AI challenge) ~ 7 animations

LAB TD3 · SOLVED

Buffers, `fork`, `exec` and `wait`

What the C library does on top of system calls, what gets copied when a process duplicates itself, and how a program launches another one and learns how it ended.

✓ **Assignment: lab report**

Hand in this lab with a short LaTeX report (just what is needed). What it must cover, the template, an example and the Prism guide are in the Assignment tab. Generative AI allowed, with a declaration.

EXERCISES

- | | | | |
|---|---|---|---|
| 1 | System call or library function? | 4 | <code>launch</code> : redirecting and running another program |
| 2 | The three stdio buffering modes | 5 | Exploration: virtual memory and the life of a process (new) |
| 3 | <code>fork</code> and the duplicated buffer | | |

TD3 · 1

● Solved

System call or library function?

Statement. Classify: `printf`, `fopen`, `fclose`, `fread`, `open`, `close`, `lseek`, `rewind`, `write`, `exit`, `_exit`. Use their prototypes and their manual section.

Solution

SYSTEM CALLS (SECTION 2)

`open`, `close`, `lseek`, `write`,
`_exit`

C LIBRARY FUNCTIONS (SECTION 3)

`printf`, `fopen`, `fclose`, `fread`, `rewind`,
`exit`

The hint is in the types: the calls work with descriptors (`int fd`); the stdio functions, with `FILE *`, a library structure that holds a descriptor **and a buffer**. `exit` flushes those buffers and runs the `atexit` functions before calling `_exit`, which ends the process at once.

TD3 · 2

● Solved · real outputs

The three stdio buffering modes

Statement. `buffer.c` writes "Hello ", sleeps one second and writes "World!". With preprocessor macros choose, for each string, `fprintf` or `write`, `stdout` or `stderr`, and at the end `exit` or `_exit`. Run each version in the terminal and through a pipe (`2>&1 | cat -u`) and identify the three buffering modes.

Solution

C tds/td3/buffer.c

```
/*
 * TD3 - exercise 2: the three buffering modes of the C library.
 * One source, several behaviours chosen at compile time (see the Makefile
 variants):
 *   STRING1, STRING2           the two strings (default "Hello " and
 "World!\n")
 *   USE_FPRINTF1/2             write string 1/2 with fprintf (buffered) instead
 of write (not buffered)
 *   USE_STDERR1/2              write string 1/2 on stderr instead of stdout
 *   USE__EXIT                  end with _exit() (no buffer flush) instead of
 exit()
 * Run each variant twice: ./buffer_N (terminal) and ./buffer_N 2>&1 |
 cat -u (pipe)
 */
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

#include "utils.h"

#ifndef STRING1
#define STRING1 "Hello "
```

```

#endif

#ifndef STRING2
#define STRING2 "World!\n"
#endif

#ifdef USE_STDERR1
#define STREAM1 stderr
#define FD1 STDERR_FILENO
#else
#define STREAM1 stdout
#define FD1 STDOUT_FILENO
#endif

#ifdef USE_STDERR2
#define STREAM2 stderr
#define FD2 STDERR_FILENO
#else
#define STREAM2 stdout
#define FD2 STDOUT_FILENO
#endif

int main(void)
{
#ifdef USE_FPRINTF1
    fputs(STRING1, STREAM1);           /* goes to the
    stdio buffer */
#else
    exit_if(write(FD1, STRING1, strlen(STRING1)) == -1, "write"); /*
    straight to the kernel */
#endif

    sleep(1);

#ifdef USE_FPRINTF2
    fputs(STRING2, STREAM2);
#else
    exit_if(write(FD2, STRING2, strlen(STRING2)) == -1, "write");
#endif

#ifdef USE__EXIT
    _exit(EXIT_SUCCESS);               /* system call: the stdio buffers are NOT
    flushed */
#else
    exit(EXIT_SUCCESS);                /* library function: flushes the stdio
    buffers, then _exit */

```

```
#endif
}
```

VERSION (MAKEFILE)	TERMINAL	PIPE	WHY
buffer_1 : fprintf(stdout) + write	World! · Hello	World! · Hello	"Hello " has no <code>\n</code> : not even line buffering flushes it; it only comes out at <code>exit</code> .
buffer_2 : write + fprintf(stderr)	Hello World!	Hello World!	<code>stderr</code> is unbuffered: every <code>fprintf</code> is an immediate write .
buffer_3 : fprintf(stdout) + fprintf(stderr)	World! · Hello	World! · Hello	The error stream goes out at once; standard output waits.
buffer_4 : like 3, with "Hello\n"	Hello · World!	World! · Hello	In a terminal <code>stdout</code> is <i>line</i> buffered (the <code>\n</code> flushes it); in a pipe it is <i>fully</i> buffered (4 KiB), so it waits for <code>exit</code> .
buffer_5 : fprintf(stdout) + <code>_exit</code>	World!	World!	<code>_exit</code> does not flush the buffer: "Hello " is lost.

Real output

```
$ ./buffer_4          # en una terminal
Hello
World!
```

Real output

```
$ ./buffer_4 2>&1 | cat -u
World!
Hello
```

Real output

```
$ ./buffer_5 2>&1 | cat -u
World!
```

The three modes: **unbuffered** (`stderr`), **line buffered** (`stdout` to a terminal) and **fully buffered** (`stdout` to a file or a pipe). `write` never has a buffer visible to the program: what it hands over, the kernel already has.

Animation A4.6 · stdio buffers →

TD3 · 3

● Solved · tested

fork and the duplicated buffer

Statement. A program prints its pid **without a newline** with `printf` and duplicates itself with `fork`; the child prints its pid and its parent's. The parent must print nothing after `fork`. What happens? Propose three solutions.

Solution

```
C tds/td3/fork_pid.c

/*
 * TD3 - exercise 3: print the pid WITHOUT a newline, fork, and let only the
 * child print.
 * SOLUTION=0 the bug: "My pid is ..." appears twice (the unflushed stdio
 * buffer is copied by fork)
 * SOLUTION=1 fflush(stdout) before fork()
 * SOLUTION=2 no stdio buffer for stdout: setvbuf(stdout, NULL, _IONBF, 0)
 * SOLUTION=3 the parent leaves with _exit(): its copy of the buffer is
 * never flushed,
 * the child's copy is printed together with the child's line
 * Make builds the four variants: fork_pid_0 ... fork_pid_3.
 */
#include <stdio.h>
#include <stdlib.h>
#include <sys/wait.h>
#include <unistd.h>

#include "utils.h"

#ifndef SOLUTION
#define SOLUTION 1
#endif
```

```
int main(void)
{
    #if SOLUTION == 2
        exit_if(setvbuf(stdout, NULL, _IONBF, 0) != 0, "setvbuf");
    #endif

    printf("My pid is %d / ", getpid());    /* no '\n': stays in the buffer
    */
    #if SOLUTION == 1
        fflush(stdout);
    #endif

    pid_t pid = fork();
    exit_if(pid == -1, "fork");

    if (pid == 0) {                          /* child */
        printf("child %d, parent %d\n", getpid(), getppid());
        exit(EXIT_SUCCESS);                  /* exit flushes the child's
    buffer */
    }

    /* Parent: prints nothing after fork, but must wait for its child. */
    int status;
    exit_if(waitpid(pid, &status, 0) == -1, "waitpid");
    #if SOLUTION == 3
        _exit(EXIT_SUCCESS);                  /* do not flush the parent's
    copy */
    #else
        return EXIT_SUCCESS;                  /* returning from main =
    exit(): flushes */
    #endif
}
```

Real output

```
$ ./fork_pid_0 | cat
My pid is 356446 / child 356448, parent 356446
My pid is 356446 /
```

Real output

```
$ ./fork_pid_1 | cat
My pid is 356458 / child 356460, parent 356458
```

Real output

```
$ ./fork_pid_3 | cat
My pid is 356482 / child 356484, parent 356482
```

The problem. "My pid is .../" stayed in the `stdout` buffer, which lives in the process memory. `fork` copies the whole memory, buffer included: parent and child hold the same pending text and both flush it when they end, so it shows up twice although the parent "prints nothing". **Solutions:** (1) `fflush(stdout)` before `fork`; (2) remove the buffer with `setvbuf(..., _IONBF, 0)`; (3) let the parent end with `_exit`, which does not flush its copy: the text comes out once, from the child. Number 1 is the clearest; the general rule is *flush the buffers before fork*.

TD3 · 4

● Solved · tested

Launch : redirecting and running another program

Statement. `./launch output command arg1 arg2...` must behave like `command arg1 arg2... > output`: it writes the command line into `output`, creates a child process that redirects its standard output with `dup2` and runs the command with `execvp`, and the parent reports how it ended (`wait(2)`). Test it with a program that causes a segmentation fault.

Solution

```
C tds/td3/launch.c

/*
 * TD3 - exercise 4: ./launch output command [args...] behaves like command
 * args... > output
 * 1. writes the command line into `output`
 * 2. forks; the child redirects its stdout to `output` with dup2 and runs
 * the command (execvp)
 * 3. the parent waits and reports how the command ended (exit status or
 * signal)
 */
#include <fcntl.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
```

```

#include <sys/wait.h>
#include <unistd.h>

#include "utils.h"

int main(int argc, char *argv[])
{
    if (argc < 3) {
        fprintf(stderr, "usage: %s output command [args...]\n", argv[0]);
        return EXIT_FAILURE;
    }

    /* O_TRUNC: like "> output", previous content is discarded. 0644 = rw-r--
r-- */
    int fd = open(argv[1], O_WRONLY | O_CREAT | O_TRUNC, 0644);
    exit_if(fd == -1, argv[1]);

    for (int i = 2; i < argc; i++) {
        exit_if(write_all(fd, argv[i], strlen(argv[i])) == -1, "write");
        exit_if(write_all(fd, i < argc - 1 ? " " : "\n", 1) == -1, "write");
    }

    pid_t pid = fork();
    exit_if(pid == -1, "fork");

    if (pid == 0) {
        /* child */
        exit_if(dup2(fd, STDOUT_FILENO) == -1, "dup2");
        exit_if(close(fd) == -1, "close"); /* 1 now points to the
file */
        execvp(argv[2], argv + 2); /* only returns on
error */
        perror(argv[2]);
        _exit(127); /* like the shell:
command not found */
    }

    exit_if(close(fd) == -1, "close"); /* the parent does not
need it */

    int status;
    exit_if(waitpid(pid, &status, 0) == -1, "waitpid");
    if (WIFEXITED(status)) {
        fprintf(stderr, "launch: %s exited with status %d\n", argv[2],
WEXITSTATUS(status));
        return WEXITSTATUS(status);
    }

```

```

    }
    if (WIFSIGNALED(status)) {
        fprintf(stderr, "launch: %s killed by signal %d (%s)%s\n", argv[2],
WTERMSIG(status),
                signame(WTERMSIG(status)), WCOREDUMP(status) ? ", core
dumped" : "");
        return 128 + WTERMSIG(status);          /* shell convention */
    }
    return EXIT_FAILURE;
}

```

C tds/td3/segfault.c

```

/*
 * TD3 - exercise 4: a program that dies with SIGSEGV, to test the error path
of launch.
 * Address 0 is never mapped: the MMU raises a page fault, the kernel finds
no mapping and
 * sends SIGSEGV to the process.
 */
#include <stdio.h>
#include <stdlib.h>

#include "utils.h"

static void gen_segfault(void)
{
    volatile int *p = NULL;    /* volatile: the compiler must really do the
store */
    *p = 42;
}

int main(void)
{
    printf("about to write at address 0...\n");
    fflush(stdout);
    gen_segfault();
    return EXIT_SUCCESS;      /* never reached */
}

```

Real output

```
$ ./launch out.txt ls -1 demo; cat out.txt
launch: ls exited with status 0
ls -1 demo
a.txt
b.txt
```

Real output

```
$ ./launch out.txt ./segfault
launch: ./segfault killed by signal 11 (SIGSEGV), core dumped
[exit status 139]
```

Real output

```
$ ./launch out.txt nocommand
nocommand: No such file or directory
launch: nocommand exited with status 127
[exit status 127]
```

- `execvp` replaces the program of the process *keeping its descriptors*: that is why the redirection happens in the child, between `fork` and `exec`. That is how the shell works.
- `execvp` only returns if it fails: the next line is always an error (127 = "not found", shell convention). `_exit` is used so that the child does not flush buffers copied from the parent.
- `WIFEXITED` / `WEXITSTATUS` and `WIFSIGNALED` / `WTERMSIG` decode the status. $139 = 128 + 11$ (SIGSEGV), the same convention the shell uses in `$?`.



Fixed

The inherited version (`redirection.c`) opened the output without `O_TRUNC` (leftovers of longer previous runs remained) and ignored the result of `write` ; the statement called it `lance` .

Exploration: virtual memory and the life of a process

Statement. Four short programs to *measure* what slides 56–82 tell: (1) `zones` prints the address of the code, the data, the heap, a large `malloc`, `printf` and the stack, and the `/proc/self/maps` region that contains each one; run it several times. (2) `lazy` reserves 1 GiB and counts resident memory and page faults before and after touching half of it. (3) `cow` fills 256 MiB, forks and counts the child's faults when reading and when writing. (4) `lifecycle` shows a zombie with `ps` before and after `waitpid`, and an adopted orphan.

Solution and real outputs

C tds/td3/zones.c

```
/*
 * TD3 - exploration 1: the memory zones of a process (original slide 62,
 * updated).
 * Prints the address of the code, the data, the heap, a large malloc, a
 * library function and the stack,
 * and the region of /proc/self/maps that contains each one. Run it twice:
 * with ASLR (address space
 * layout randomization) the addresses change on every run, but the order of
 * the zones does not.
 */
#include <stdint.h>
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

#include "utils.h"

int global_array[10];           /* uninitialized global: .bss */
static int initialized = 42;    /* initialized global: .data */

/* Finds the line of /proc/self/maps whose range contains addr and prints its
 * permissions and name. */
static void show(const char *what, const void *addr)
{
    FILE *maps = fopen("/proc/self/maps", "r");
    exit_if(maps == NULL, "/proc/self/maps");

    char line[512];
```

```

uintptr_t a = (uintptr_t)addr;
while (fgets(line, sizeof line, maps) != NULL) {
    unsigned long start, end;
    char perms[5], path[256] = "";
    if (sscanf(line, "%lx-%lx %4s %*s %*s %*s %255[^\n]", &start, &end,
perms, path) < 3)
        continue;
    if (a ≥ start && a < end) {
        char *name = path;
        while (*name == ' ')
            name++;
        printf("%-24s %#14lx  %s  %s\n", what, (unsigned long)a, perms,
*name ? name : "(anonymous)");
        break;
    }
}
exit_if(fclose(maps) == EOF, "fclose");
}

int main(void)
{
    int local = 0;
    void *small = malloc(10);
    void *big = malloc(64 << 20);          /* above M_MMAP_THRESHOLD (128 KiB):
glibc uses mmap */
    exit_if(small == NULL || big == NULL, "malloc");

    printf("%-24s %14s  %s  %s\n", "zone", "address", "perm", "region");
    show("code   (main)", (const void *)main);
    show("data   (initialized)", &initialized);
    show("bss    (global_array)", global_array);
    show("heap   (malloc 10 B)", small);
    show("heap   (end: sbrk(0))", (char *)sbrk(0) - 1);
    show("mmap   (malloc 64 MiB)", big);
    show("libc   (printf)", (const void *)printf);
    show("stack  (local)", &local);

    free(big);
    free(small);
    return EXIT_SUCCESS;
}

```

Real output

```
$ ./zones
zone                address    perm  region
code  (main)         0x5f935f9d1350  r-xp  ~/fso/tds/bin/td3/zones
data  (initialized)    0x5f935f9d5010  rw-p  ~/fso/tds/bin/td3/zones
bss   (global_array)   0x5f935f9d5040  rw-p  ~/fso/tds/bin/td3/zones
heap  (malloc 10 B)     0x5f938dc96010  rw-p  [heap]
heap  (end: sbrk(0))   0x5f938dcb6fff  rw-p  [heap]
mmap  (malloc 64 MiB)  0x71b340fff010  rw-p  (anonymous)
libc  (printf)         0x71b345064080  r-xp  /usr/lib/x86_64-linux-
gnu/libc.so.6
stack (local)         0x7fff6c1e68b4  rw-p  [stack]
```

Real output

```
$ for i in 1 2 3; do ./zones | grep main; done
code  (main)         0x57a1f784a350  r-xp  ~/fso/tds/bin/td3/zones
code  (main)         0x5d648e064350  r-xp  ~/fso/tds/bin/td3/zones
code  (main)         0x56e46bc21350  r-xp  ~/fso/tds/bin/td3/zones
```

The order of the zones stays (code, data and heap at the bottom; libraries and `mmap` above; the stack at the very top), but the base changes on every run: **ASLR**. The 64 MiB `malloc` is not in the heap: glibc uses `mmap` above 128 KiB.

C tds/td3/lazy.c

```
/*
 * TD3 - exploration 2: lazy allocation (original slides 59 and 61).
 * malloc of a large block returns at once and costs almost no physical
 * memory: the kernel only
 * reserves virtual addresses. Each page gets a physical frame the first time
 * it is touched, through a
 * (minor) page fault. The program measures the resident memory (VmRSS) and
 * the page faults
 * (getrusage) before and after touching half of the block.
 * ./lazy [MiB]          default: 1024 MiB
 */
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/resource.h>
#include <time.h>
#include <unistd.h>
```

```
#include "utils.h"

static long minor_faults(void)
{
    struct rusage ru;
    exit_if(getrusage(RUSAGE_SELF, &ru) == -1, "getrusage");
    return ru.ru_minflt;
}

/* Resident set size in KiB, from /proc/self/status. */
static long rss_kib(void)
{
    FILE *f = fopen("/proc/self/status", "r");
    exit_if(f == NULL, "/proc/self/status");
    char line[256];
    long kib = -1;
    while (fgets(line, sizeof line, f) != NULL)
        if (sscanf(line, "VmRSS: %ld kB", &kib) == 1)
            break;
    exit_if(fclose(f) == EOF, "fclose");
    return kib;
}

static double now(void)
{
    struct timespec ts;
    exit_if(clock_gettime(CLOCK_MONOTONIC, &ts) == -1, "clock_gettime");
    return ts.tv_sec + ts.tv_nsec / 1e9;
}

static void report(const char *step, long f0)
{
    printf("%-34s VmRSS = %8ld KiB    page faults so far = %ld\n", step,
    rss_kib(), minor_faults() - f0);
}

int main(int argc, char *argv[])
{
    size_t mib = argc > 1 ? (size_t)atol(argv[1]) : 1024;
    size_t size = mib << 20, page = (size_t)sysconf(_SC_PAGESIZE);
    long f0 = minor_faults();

    report("start", f0);
    char *p = malloc(size);
    exit_if(p == NULL, "malloc");
}
```

```
printf("malloc(%zu MiB) returned %p\n", mib, (void *)p);
report("after malloc", f0);

long before = minor_faults();
double t0 = now();
for (size_t i = 0; i < size / 2; i += page)    /* touch one byte of
every page of the first half */
    p[i] = 1;
double t1 = now();
long faults = minor_faults() - before;
report("after touching half of the pages", f0);
printf("pages touched: %zu (%zu bytes each)    page faults: %ld    %.0f ns
per fault\n",
        size / 2 / page, page, faults, faults ? (t1 - t0) * 1e9 / faults :
0.0);

free(p);
report("after free", f0);
return EXIT_SUCCESS;
}
```

Real output

```
$ ./lazy 1024
start                                VmRSS =    1756 KiB    page faults so far
= 0
malloc(1024 MiB) returned 0x7fadb67ff010
after malloc                          VmRSS =    1952 KiB    page faults so far
= 8
after touching half of the pages    VmRSS =  526300 KiB    page faults so far
= 131081
pages touched: 131072 (4096 bytes each)    page faults: 131073    1968 ns per
fault
after free                            VmRSS =    2012 KiB    page faults so far
= 131081
```

`malloc(1 GiB)` does not change resident memory: it only reserves addresses. Every page touched for the first time causes a page fault in which the kernel allocates and zeroes a frame: as many faults as pages.

C tds/td3/cow.c

```
/*
 * TD3 - exploration 3: copy-on-write after fork (original slide 74).
 * The parent fills a block, so all its pages are resident. After fork,
```

```

parent and child share the
    * physical pages, marked read-only. The child reads the whole block (no
copy: no page faults) and then
    * writes to a quarter of the pages: each first write causes a page fault in
which the kernel copies
    * that single page. The page faults are counted with getrusage.
    * ./cow [MiB]          default: 256 MiB
    */
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/resource.h>
#include <sys/wait.h>
#include <time.h>
#include <unistd.h>

#include "utils.h"

static long minor_faults(void)
{
    struct rusage ru;
    exit_if(getrusage(RUSAGE_SELF, &ru) == -1, "getrusage");
    return ru.ru_minflt;
}

static double now(void)
{
    struct timespec ts;
    exit_if(clock_gettime(CLOCK_MONOTONIC, &ts) == -1, "clock_gettime");
    return ts.tv_sec + ts.tv_nsec / 1e9;
}

int main(int argc, char *argv[])
{
    size_t mib = argc > 1 ? (size_t)atol(argv[1]) : 256;
    size_t size = mib << 20, page = (size_t)sysconf(_SC_PAGESIZE), pages =
size / page;
    char *p = malloc(size);
    exit_if(p == NULL, "malloc");
    memset(p, 'x', size);                                /* every page is
now resident */
    printf("parent: %zu MiB filled (%zu pages of %zu bytes)\n", mib, pages,
page);
    fflush(stdout);                                       /* do not
duplicate the stdio buffer */

```

```
double t0 = now();
pid_t pid = fork();
exit_if(pid == -1, "fork");
if (pid == 0) {
    printf("fork took %.2f ms (only the page tables are copied)\n",
(now() - t0) * 1e3);

    long f = minor_faults();
    unsigned long sum = 0;
    for (size_t i = 0; i < size; i += page)
        sum += (unsigned char)p[i];
    printf("child reads %zu pages: %ld page faults (shared, nothing
copied; sum %lu)\n", pages, minor_faults() - f, sum);

    f = minor_faults();
    double t1 = now();
    for (size_t i = 0; i < size / 4; i += page)
        p[i] = 'y';
    double t2 = now();
    long faults = minor_faults() - f;
    printf("child writes %zu pages: %ld page faults (one copy per page),
%.0f ns per copy\n",
        pages / 4, faults, faults ? (t2 - t1) * 1e9 / faults : 0.0);
    free(p);
    exit(EXIT_SUCCESS);
}

int status;
exit_if(waitpid(pid, &status, 0) == -1, "waitpid");
printf("parent: the child exited with status %d; my copy still says
'%c'\n", WEXITSTATUS(status), p[0]);
free(p);
return EXIT_SUCCESS;
}
```

Real output

```
$ ./cow 256
parent: 256 MiB filled (65536 pages of 4096 bytes)
fork took 6.70 ms (only the page tables are copied)
child reads 65536 pages: 0 page faults (shared, nothing copied; sum 7864320)
child writes 16384 pages: 16384 page faults (one copy per page), 2753 ns per
copy
parent: the child exited with status 0; my copy still says 'x'
```

After `fork`, reading copies nothing (0 faults); writing causes exactly one fault—one copy— per page written. That is why forking a 256 MiB process takes milliseconds: it only copies the page tables.

```
C tds/td3/lifecycle.c

/*
 * TD3 - exploration 4: the end of a process: zombies and orphans (original
 * slides 80-82 and 88).
 * 1. A child that exits stays as a ZOMBIE (state Z, <defunct>) until its
 * parent calls wait: the
 *     kernel keeps its PID and exit status. ps shows it before and after
 * waitpid.
 * 2. A process whose parent dies is an ORPHAN: the kernel gives it a new
 * parent, init (PID 1) or the
 *     nearest ancestor marked as "subreaper" (on a desktop, usually systemd
 * --user).
 */
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <sys/wait.h>
#include <time.h>
#include <unistd.h>

#include "utils.h"

static void pause_ms(long ms)
{
    struct timespec ts = { ms / 1000, (ms % 1000) * 1000000L };
    while (nanosleep(&ts, &ts) == -1)
        ;
}

static void ps(pid_t pid)
{
    char cmd[128];
    snprintf(cmd, sizeof cmd, "ps -o pid,ppid,stat,comm -p %d | sed 's/^/'", (int)pid);
    fflush(stdout);
    if (system(cmd) == -1)
        perror("system");
}

/* Name of a process, from /proc/PID/comm. */
static void comm_of(pid_t pid, char *name, size_t len)
```

```

{
    char path[64];
    snprintf(path, sizeof path, "/proc/%d/comm", (int)pid);
    FILE *f = fopen(path, "r");
    if (f == NULL || fgets(name, (int)len, f) == NULL)
        snprintf(name, len, "?");
    name[strcspn(name, "\n")] = '\0';
    if (f != NULL)
        fclose(f);
}

int main(void)
{
    /* 1. Zombie */
    pid_t child = fork();
    exit_if(child == -1, "fork");
    if (child == 0)
        _exit(7);
    pause_ms(200);
    printf("1) the child %d has exited, but nobody has called wait yet:\n",
(int)child);
    ps(child);
    int status;
    exit_if(waitpid(child, &status, 0) == -1, "waitpid");
    printf("    waitpid → WIFEXITED = %d, WEXITSTATUS = %d; now ps finds
nothing:\n", WIFEXITED(status), WEXITSTATUS(status));
    ps(child);

    /* 2. Orphan: the middle process exits, the grandchild is adopted */
    printf("2) orphan: my PID is %d\n", (int)getpid());
    fflush(stdout);
    pid_t middle = fork();
    exit_if(middle == -1, "fork");
    if (middle == 0) {
        pid_t grandchild = fork();
        exit_if(grandchild == -1, "fork");
        if (grandchild == 0) {
            pid_t before = getppid();
            pause_ms(200);
            /* meanwhile its parent
exits */
            pid_t after = getppid();
            char name[64];
            comm_of(after, name, sizeof name);
            printf("    grandchild %d: parent was %d, now it is %d (%s)\n",
(int)getpid(), (int)before, (int)after, name);

```

```
        exit(EXIT_SUCCESS);
    }
    exit(EXIT_SUCCESS);                /* the grandchild becomes
an orphan */
    }
    exit_if(waitpid(middle, NULL, 0) == -1, "waitpid");
    pause_ms(400);                    /* let the grandchild
report */
    return EXIT_SUCCESS;
}
```

Real output

```
$ ./lifecycle
1) the child 356667 has exited, but nobody has called wait yet:
    PID    PPID STAT COMMAND
    356667 356666 Z    lifecycle
waitpid → WIFEXITED = 1, WEXITSTATUS = 7; now ps finds nothing:
    PID    PPID STAT COMMAND
2) orphan: my PID is 356666
grandchild 356679: parent was 9771, now it is 9771 (systemd)
```

The finished child shows state `Z` until `waitpid`, which collects its code (7). The orphaned grandchild does not go to `init` but to the user session's `systemd`, a *subreaper* (`prctl(PR_SET_CHILD_SUBREAPER)`).

See it animated → [A4.1 virtual memory](#) · [A4.2 copy-on-write](#) · [A4.3 states](#)

Checklist

Generative AI challenge · A mini shell



Goal. Turn `launch` into a minimal command interpreter: it reads lines, runs commands with `fork/execvp`, supports `> file` and shows the exit status.

Starting prompt. Use it in the tool you prefer (Claude, ChatGPT, Gemini, Copilot, Codex...) with the *solved* version of the lab as context. Ask for a plan first and review every step before accepting it.

Prompt

Starting from my `launch.c` [paste it], help me write a mini shell in C that: reads lines with `getline`, splits them into words, runs the command with `fork + execvp`, supports `> file` redirection with `dup2`, implements `cd` and `exit` as built-ins (explain why `cd` cannot be an external program) and prints `[status N]` or `[signal SIGxxx]` when a command ends. Ask me to write each part before showing me yours.

How to check the result

- `ls -l > out.txt` creates the file and the shell keeps working.
- `cd /tmp` followed by `pwd` prints `/tmp`.
- Running `./segfault` shows the signal and leaves no zombie processes (`ps`).

Declare your use of AI: tool, relevant prompts, what you checked yourself and what you fixed. What counts is that you can explain and verify every decision.